
FOUNDATION & ENVIRONMENT SETUP REPORT

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Domain: Cybersecurity Internship

1. Objective & Scope

The primary goal was to architect a controlled **Sandboxed Environment** for ethical hacking. This involves a "Victim-Attacker" model where security vulnerabilities can be studied without affecting real-world infrastructure. The environment is designed to be completely isolated from the host machine and external networks to ensure safety and ethical compliance.

2. Lab Architecture & Virtual Machine Profiles

I utilized a **Host-Only Networking** strategy. Unlike NAT or Bridged modes, Host-Only creates a dedicated virtual switch disconnected from the physical network card, preventing the vulnerable target from being exposed to the public internet.

2.1 Attacker Machine: Kali Linux

Kali Linux is a specialized Debian-based distribution pre-loaded with penetration testing tools. It serves as the "Command Center" for this lab.

- **Operating System:** Kali GNU/Linux
- **Configured IP:** 192.168.56.102
- **Role:** Reconnaissance, vulnerability scanning, and traffic interception.

2.2 Target Machine: Metasploitable2

Metasploitable2 is an intentionally vulnerable Ubuntu-based VM. It is purposely misconfigured with outdated services to provide a wide attack surface for training.

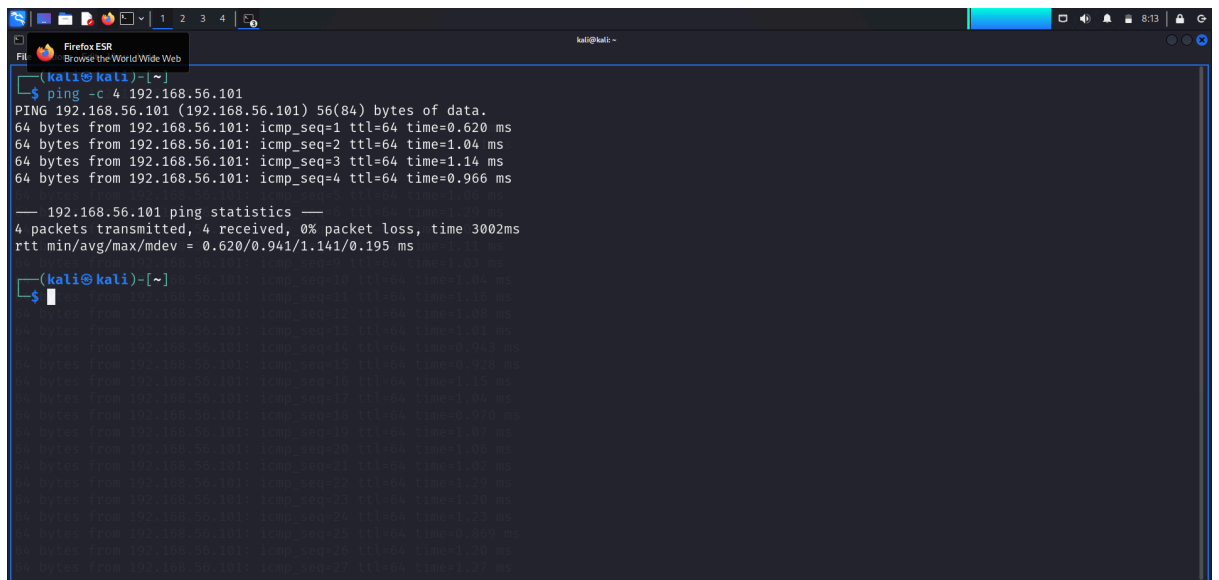
- **Operating System:** Ubuntu 8.04
- **Configured IP:** 192.168.56.101
- **Role:** Victim machine for vulnerability identification and exploit testing.

3. Connectivity & Protocol Analysis

3.1 ICMP Connectivity (Ping Test)

Connectivity was verified using the **Internet Control Message Protocol (ICMP)**. By sending an **Echo Request** from Kali to Metasploitable, I confirmed that the Network Layer (Layer 3) was correctly configured and responding.

- **Command:** `ping -c 4 192.168.56.101`
- **Result:** 0% packet loss, confirming a stable link.

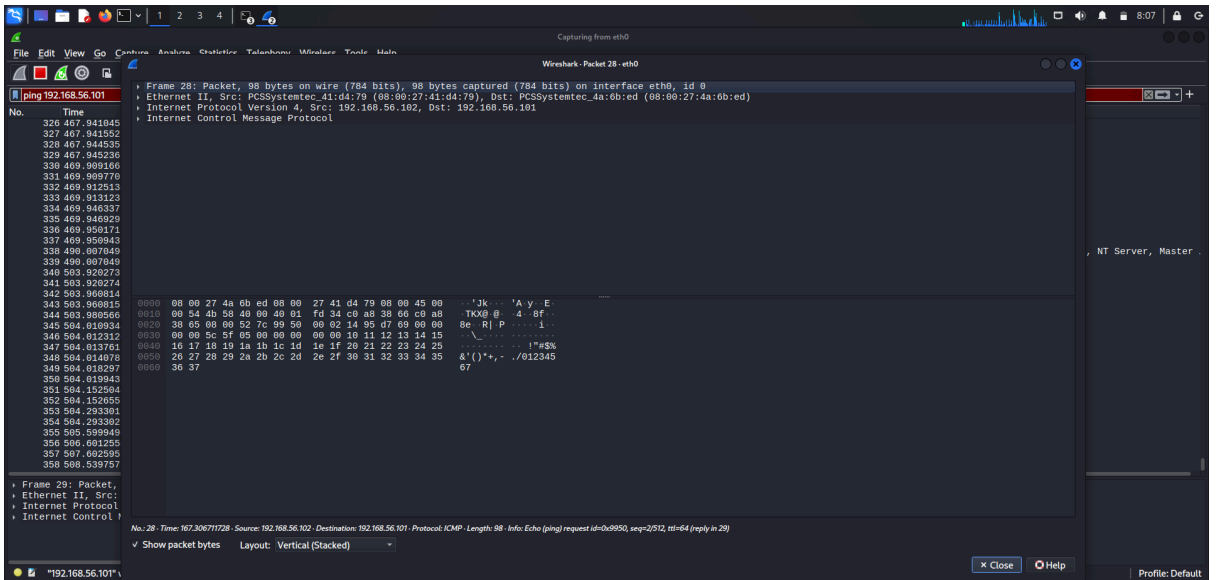
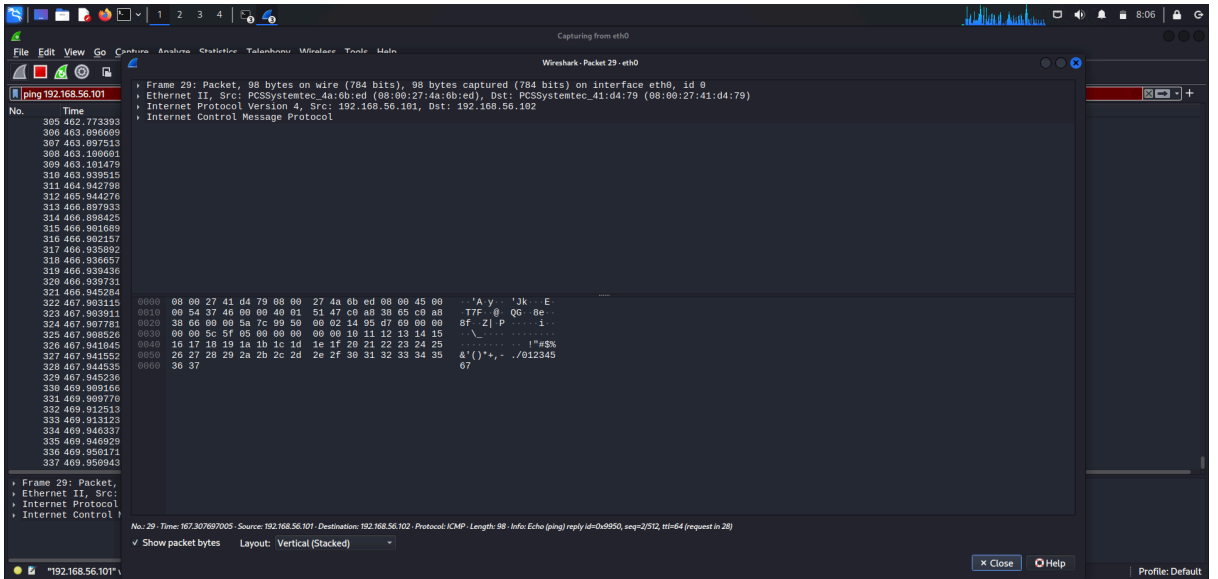


```
(kali㉿kali)-[~]
$ ping -c 4 192.168.56.101
PING 192.168.56.101 (192.168.56.101) 56(84) bytes of data:
64 bytes from 192.168.56.101: icmp_seq=1 ttl=64 time=0.620 ms
64 bytes from 192.168.56.101: icmp_seq=2 ttl=64 time=1.04 ms
64 bytes from 192.168.56.101: icmp_seq=3 ttl=64 time=1.14 ms
64 bytes from 192.168.56.101: icmp_seq=4 ttl=64 time=0.966 ms
— 192.168.56.101 ping statistics —
4 packets transmitted, 4 received, 0% packet loss, time 3002ms
rtt min/avg/max/mdev = 0.620/0.941/1.141/0.195 ms
(kali㉿kali)-[~]
$
```

3.2 Packet Capture Analysis (Wireshark)

Wireshark was used to perform **Deep Packet Inspection (DPI)**. I captured the live traffic on the **eth0** interface to verify the flow of data.

- **Technical Insight:** I verified the **ARP** requests mapping IP addresses to virtual MAC addresses, followed by the ICMP flow, confirming that Layers 2 and 3 of the OSI model are functional.



4. Advanced Tool Implementation

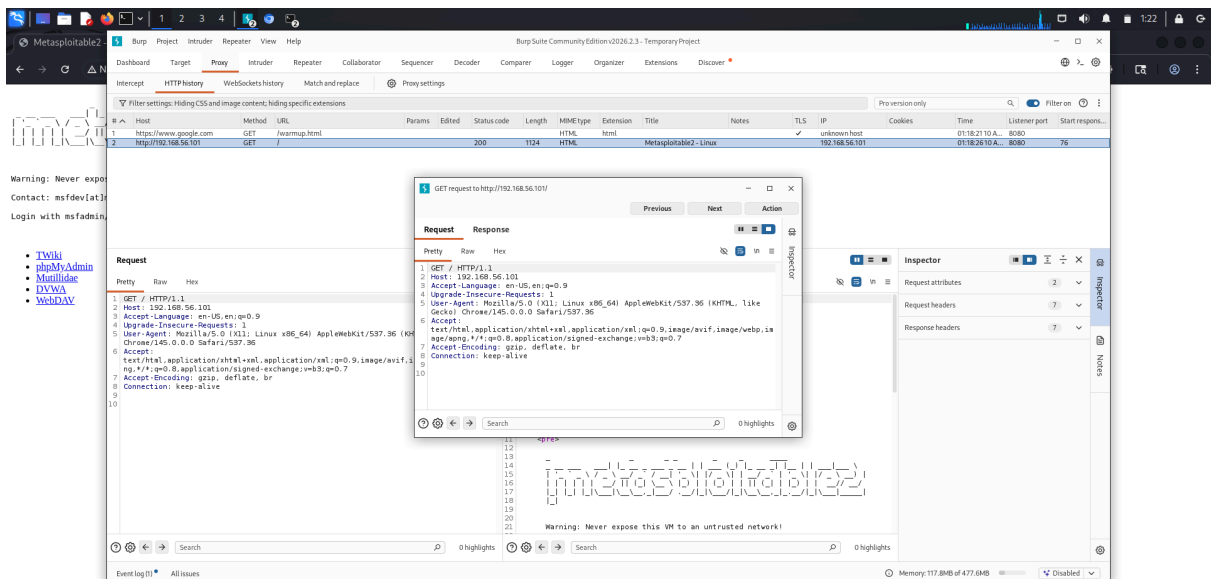
A. Network Mapping (Nmap)

Nmap was used for **Service Version Detection (-sV)**. This allowed me to identify the exact software versions running on open ports (e.g., Apache 2.2.8, vsftpd 2.3.4), which is the first step in identifying specific CVEs.

```
kali@kali:~$ nmap -Pn -sV 192.168.56.101
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-04-09 07:56 EDT
Nmap scan report for 192.168.56.101
Host is up (0.0027s latency).
Not shown: 977 closed tcp ports (conn-refused)
PORT      STATE SERVICE        VERSION
21/tcp    open  ftp            vsftpd 2.3.4
22/tcp    open  ssh            OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp    open  telnet        Linux telnetd
25/tcp    open  smtp          Postfix smtpd
53/tcp    open  domain        ISC BIND 9.4.2
80/tcp    open  http          Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp   open  rpcbind       2 (RPC #100000)
139/tcp   open  netbios-ssn   Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn   Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp   open  exec          netkit-rsh rexecd
513/tcp   open  login         OpenBSD or Solaris rlogind
514/tcp   open  shell         Netkit rshd
1099/tcp  open  java-rmi      GNU Classpath grmiregistry
1524/tcp  open  bindshell     Metasploitable root shell
2049/tcp  open  nfs           2-4 (RPC #100003)
2121/tcp  open  ftp          ProFTPD 1.3.1
3306/tcp  open  mysql         MySQL 5.0.51a-3ubuntu5
5432/tcp  open  postgresql    PostgreSQL DB 8.3.0 - 8.3.7
5900/tcp  open  vnc           VNC (protocol 3.3)
6000/tcp  open  X11           (access denied)
6667/tcp  open  irc           UnrealIRCd
8009/tcp  open  ajp13         Apache Jserv (Protocol v1.3)
8180/tcp  open  http          Apache Tomcat/Coyote JSP engine 1.1
```

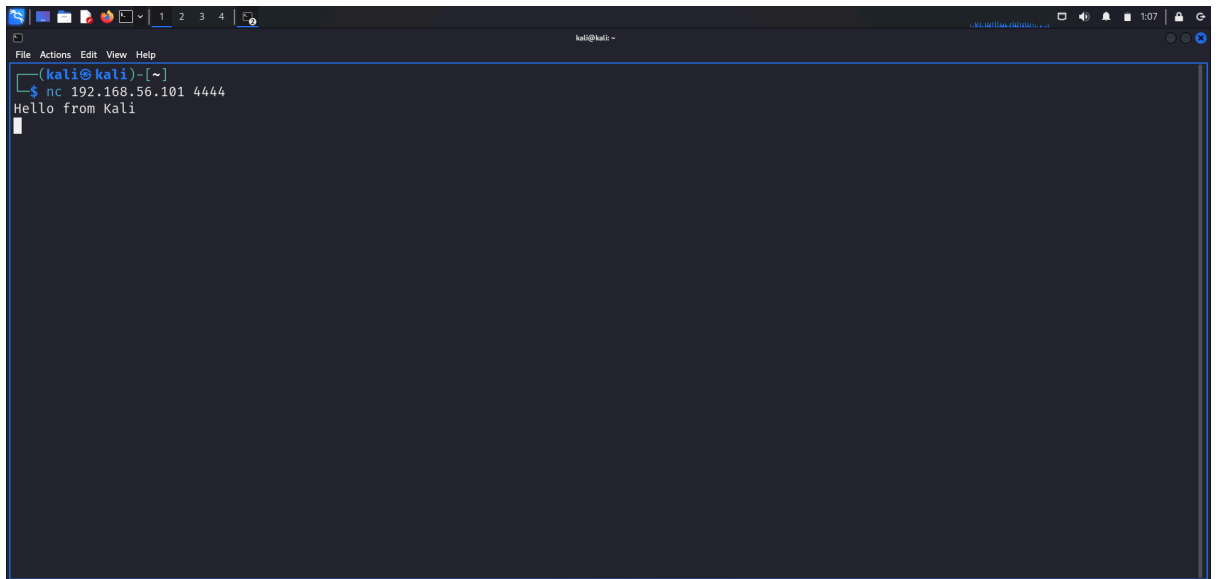
B. Intercepting Proxy (Burp Suite)

Burp Suite was implemented as a **Man-in-the-Middle (MITM)** proxy. By intercepting the **HTTP Request**, I can view and manipulate sensitive data such as Cookies and Headers before they reach the server.



C. Network Debugging (Netcat)

Netcat was used to demonstrate **Raw Socket Programming**. By opening a listener on the target and a connector on the attacker, I proved that manual data streams can be established bypassing traditional application layers.



```
(kali㉿kali)-[~]  
$ nc 192.168.56.101 4444  
Hello from Kali
```

```
Warning: Never expose this VM to an untrusted network!  
  
Contact: msfdev[at]metasploit.com  
  
Login with msfadmin/msfadmin to get started  
  
metasploitable login: msfadmin  
Password:  
Last login: Thu Apr  9 07:52:45 EDT 2026 on tty1  
Linux metasploitable 2.6.24-16-server #1 SMP Thu Apr 10 13:58:00 UTC 2008 i686  
  
The programs included with the Ubuntu system are free software;  
the exact distribution terms for each program are described in the  
individual files in /usr/share/doc/*/copyright.  
  
Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by  
applicable law.  
  
To access official Ubuntu documentation, please visit:  
http://help.ubuntu.com/  
No mail.  
msfadmin@metasploitable:~$ nc -lp 4444  
Hello from Kali
```

5. Cybersecurity Theoretical Framework

5.1 The CIA Triad

- **Confidentiality:** Handled by the private 192.168.56.x range, which is not routable over the internet.
- **Integrity:** Verified through Wireshark to ensure packets arrived unaltered.
- **Availability:** Demonstrated by the stable uptime of the virtualized services during testing.

5.2 Cryptography Demonstration

I used **OpenSSL** on Kali to demonstrate symmetric encryption.

- **Command:** `openssl enc -aes-256-cbc -salt -in secret.txt -out secret.enc`

6. Conclusion

The lab environment is successfully deployed and secured. I have verified connectivity at the Network Layer (Ping), the Transport Layer (Netcat), and the Application Layer (Burp Suite).
